

Energy storage in inductors

$$p = vi$$

$$w = \int v i dt = \int L \frac{di}{dt} i dt = L \int i di = \frac{1}{2} L i^2$$

$$\boxed{w = \frac{1}{2} L i^2}$$
 is the stored energy in an inductor, in joules.

Series and parallel inductors

Series inductors have the same current.

$$v = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + \dots + L_n \frac{di}{dt} = (L_1 + L_2 + \dots + L_n) \frac{di}{dt} = L_{eq} \frac{di}{dt}$$

$$L_{eq} = L_1 + L_2 + \dots + L_n$$

Series inductors add like series resistors.

Parallel inductors add like parallel resistors.

$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2} \quad L_{eq} = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_n}}$$

The real fast way to solve RC circuits (the cookbook)

If you think about it, ANY circuit with one capacitor can be reduced to the equivalent circuit with one resistor and one voltage source just by taking the Thévenin equivalent seen by the capacitor. Which means that ANY circuit with one capacitor has a solution of the form

$$v(t) = v_{oc} + (v(0+) - v_{oc}) e^{-t/\tau}$$

Let's rewrite this slightly: $v(t) = A + B e^{-t/\tau}$

Of course, A and B can be found from v_{oc} and $v(0+)$, but you don't really need to remember that. It's easier to find A and B from $v(0+)$ and $v(\infty)$, doing a little algebra instead of memorizing an equation.

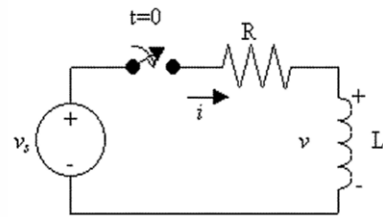
RL circuits-3

An astute observer will note that the right hand side (the forcing function in mathematical terms) is the Norton equivalent short circuit current. Contrast this to the RC circuit where we used the Thévenin equivalent open circuit voltage.

Thus, the form of the solution is

$$i(t) = A + Be^{-t/\tau}$$

where the time constant $\tau = L/R$



时间常数（Time Constant），通常用希腊字母 τ (tau) 表示，是指电路对状态改变的“反应速度”

Writing the differential equation-5

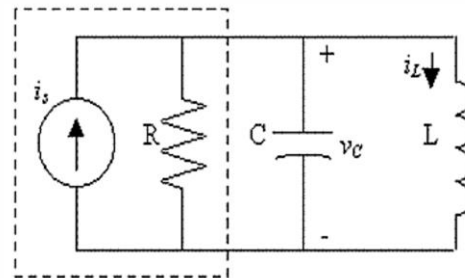
And from the inductor branch relationship

$$v_C = L \frac{di_L}{dt} \quad \text{which is} \quad x_1 = f\left(\frac{dx_2}{dt}\right)$$

$$\frac{d}{dt}\left(L \frac{di_L}{dt}\right) = -\frac{1}{RC}\left(L \frac{di_L}{dt}\right) - \frac{i_L}{C} + \frac{i_s}{C}$$

and rearrange:

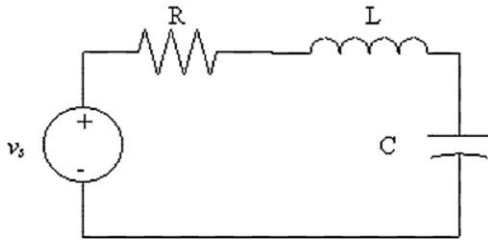
$$LC \frac{d^2 i_L}{dt^2} + \frac{L}{R} \frac{di_L}{dt} + i_L = i_s,$$



which is the second order differential equation for the inductor current.

Writing the differential equation-6

Now you try it for a series RLC circuit:



Choose your state variables:

Now write KVL around the loop:
And the branch relationship for the capacitor:

Substitute in for inductor current,
and you should end up with
something like:

$$LC \frac{d^2 v_C}{dt^2} + RC \frac{dv_C}{dt} + v_C = v_s$$

3. 如何计算 α 和 ω_o ?

对于不同的电路结构，计算公式不同，这是最容易出错的地方：

- 并联 RLC:

- $\alpha = \frac{1}{2RC}$
 - $\omega_o = \frac{1}{\sqrt{LC}}$

- 串联 RLC:

- $\alpha = \frac{R}{2L}$
 - $\omega_o = \frac{1}{\sqrt{LC}}$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_o^2}$$

① 过阻尼 ($\alpha > \omega_o$)

特征根为 $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_o^2}$ 。

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

② 临界阻尼 ($\alpha = \omega_o$)

特征根为 $s_1 = s_2$ 。

$$v(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t}$$

③ 欠阻尼 ($\alpha < \omega_o$)

引入阻尼振荡频率 $\omega_d = \sqrt{\omega_o^2 - \alpha^2}$ 。

$$v(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

1. 无阻尼 (Undamped)

- 条件: $R \rightarrow \infty$ (对于并联电路) 或 $R = 0$ (对于串联电路)。简单说就是电路中没有电阻。
- 根的特点: 两个纯虚根 $s = \pm j\omega_0$ 。
- 图像特点: 等幅振荡。能量在 L 和 C 之间永无止境地交换, 波形是完美的正弦波, 振幅永远不减小。

2. 欠阻尼 (Underdamped)

- 条件: 电阻值较大 (并联) 或较小 (串联), 满足阻尼系数小于振荡频率。
- 根的特点: 一对共轭复根 $s = -\alpha \pm j\omega_d$ 。
- 图像特点: 衰减振荡。电路会像被拨动的琴弦一样来回抖动, 但由于电阻在耗能, 每一次抖动的幅度都比上一次小, 最终趋于零。

3. 临界阻尼 (Critically Damped)

- 条件: 刚好处于振荡与不振荡的分界点。
- 根的特点: 两个相等的实根 $s_1 = s_2$ 。
- 图像特点: 最快稳定且无振荡。它没有来回的抖动, 是电路以最短时间达到最终稳定状态的“完美平衡点”。

4. 过阻尼 (Overdamped)

- 条件: 电阻非常小 (并联) 或非常大 (串联), 阻力极大。
- 根的特点: 两个不相等的负实根 $s_1 \neq s_2$ 。
- 图像特点: 缓慢爬升/下降。因为阻力太大, 电荷移动非常吃力, 电路像掉进了厚重的蜂蜜里, 需要很长时间才能磨蹭到稳定值。

Underdamped case

This is the complex conjugate case, $s = \alpha \pm j\omega$

Plugging in to the two-root solution form,

$$i_n = A_1 e^{(\alpha + j\omega)t} + A_2 e^{(\alpha - j\omega)t} = A_1 e^{\alpha t} e^{j\omega t} + A_2 e^{\alpha t} e^{-j\omega t}$$

$$i_n = e^{\alpha t} (A_1 e^{j\omega t} + A_2 e^{-j\omega t})$$



Heinrich Rudolph Hertz (1857-1894)*

And we should be able to recognize the second factor as the exponential way of expressing a sinusoid, so:

$$i_n = e^{\alpha t} (B_1 \cos \omega t + B_2 \sin \omega t)$$

Which is the form of the solution for the underdamped case

Yes, it is just as you suspected! ω is a radial velocity (also known as radian frequency) with units of radians per second, and $\omega = 2\pi f$, where f is the frequency in Hertz (cycles per second), abbreviated Hz.

$$B_1 = A_1 + A_2$$

$$B_2 = j(A_1 - A_2)$$

9.2 正弦响应

在研究正弦信号源的稳态响应之前,先从大的范围讨论一下这个问题,首先看电路的全响应,这将有助于理解电路的稳态响应。电路如图 9.5 所示,其中 v_s 为一正弦电压,即

$$v_s = V_m \cos(\omega t + \phi) \quad (9.7)$$

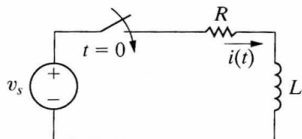


图 9.5 正弦电压源激励下的 RL 电路

为方便起见,假设电路中的初始电流为 0,推导出 $t \geq 0$ 时的 $i(t)$ 表达式。这类似于第 7 章中所求 RL 电路的阶跃响应,唯一的区别在于这里的电压源是随时间变化的正弦电压,而不是常数,或者说不是直流电压。对图 9.5 直接利用基尔霍夫电压定律,得到微分方程式 (9.8):

$$L \frac{di}{dt} + Ri = V_m \cos(\omega t + \phi) \quad (9.8)$$

此式的解在有关微分式的课程中讨论过,可直接使用结论,得到 i 的解为:

$$i = \frac{-V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\phi - \theta) e^{-(R/L)t} + \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\omega t + \phi - \theta) \quad (9.9)$$

其中 $\theta = \omega L/R$,若正弦信号源的频率已知,即可方便地求出电路的初相角 θ 。

为了验证式 (9.9) 的正确性,可以将其代入到式 (9.8) 中,看看对于 $t \geq 0$ 是否满足,在习题 9.10 中将此留做练习,让读者自己证明。

式 (9.9) 右边的第一个分量称为电流的暂态分量,因为随着时间的推移,它将趋于无穷小;第二个分量称为稳态分量,只要开关保持闭合状态且信号源为正弦电压,这个分量就一直存在,暂态分量响应和稳态分量响应合称为全响应。在这一章里,避开微分式的求解方法,直接讨论稳态响应的求解方法,将有助于求解暂态响应和全响应。

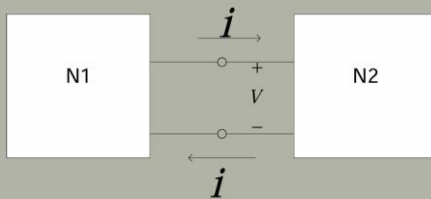
研究式 (9.9) 中的稳态分量,得到下面几条关于稳态响应的性质。

1. 稳态响应为一正弦函数。
2. 响应频率与信号源频率相同,这一性质适用于电路参数 R, L, C 均为常数的任一线性电路 (如果响应频率与信号源频率不同,则电路中含有非线性元件)。
3. 稳态响应的最大值通常与信号源的幅值不同,以上面讨论的电路为例,响应的最大幅值为 $V_m / \sqrt{R^2 + \omega^2 L^2}$,而信号源的幅值为 V_m 。
4. 稳态响应的相位角通常与信号源不同,以上面讨论的电路为例,电流的相位角为 $\phi - \theta$,而电压源的相位角是 ϕ 。

记住这些性质,有利于理解 9.3 节中介绍的相量求解法。求电路的稳态响应实际上是求响应的幅值及相位角,因为响应的形状及其频率已知。

注意 θ 的表达式,是 $\theta = \arctan(\omega L/R)$

Instantaneous Power



In sinusoidal steady-state, the voltage and current will generally be

$$v(t) = V_m \cos(\omega t + \theta_v); i(t) = I_m \cos(\omega t + \theta_i)$$

Thus:

$$p(t) = v(t)i(t) = I_m V_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

Some useful trig identities

$$\begin{aligned} \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ \cos^2 \alpha &= \frac{1}{2} + \frac{1}{2} \cos 2\alpha; \sin^2 \alpha = \frac{1}{2} - \frac{1}{2} \cos 2\alpha; \cos \alpha \sin \alpha = \frac{1}{2} \sin 2\alpha \\ \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \end{aligned}$$

$$\begin{aligned} p(t) &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \cos 2\omega t \\ &\quad - \frac{1}{2} V_m I_m \sin(\theta_v - \theta_i) \sin 2\omega t \end{aligned}$$

Instantaneous Power

Let $P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$ average power[Watt]

$Q = \frac{1}{2} V_m I_m \sin(\theta_v - \theta_i)$ reactive power[VAR]

This yields,

$$p(t) = P + P \cos 2\omega t - Q \sin 2\omega t$$

通过积分可知，P 就是平均功率

先求出支路电流的有效值 I_{rms} 。

求有功 Active Power: 直接找电阻， $P = \sum I^2 R$ 。

求无功 Reactive Power: 直接找电感和电容， $Q = \sum I^2 X$ 。

注意 无功功率是乘以 X，也就是说在计算电容的时候，会有一个负号

求视在功率 Apparent Power: $S = \sqrt{P^2 + Q^2}$ ，或者直接 $S = I_{rms}^2 |Z|$ 。

不能直接使用 U^2/R 或者 X 来计算，二者都会分压

已知条件	复功率 S 公式	得到的 P (实部)	得到的 Q (虚部)
电压 V、电流 I	$V \cdot I^*$	$VI \cos(\theta_v - \theta_i)$	$VI \sin(\theta_v - \theta_i)$
电流 I、阻抗 Z	$I^2 Z$	$I^2 R$	$I^2 X$
电压 U、阻抗 Z	U^2 / Z^*	$U^2 \cdot \frac{R}{ Z ^2}$	Z
视在功率 S、角度 θ	$S \angle \theta$	$S \cos \theta$	$S \sin \theta$

Power for purely resistive circuit

$$p(t) = P + P \cos 2\omega t, \text{ where } \theta_v = \theta_i \quad P = \frac{1}{2} V_m I_m$$

$$Q = 0$$

Electric energy is dissipated in the form of thermal energy

Power for purely inductive circuit

$$p(t) = -Q \sin 2\omega t \text{ where } \theta_v = \theta_i + 90^\circ$$

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = 0$$

$$Q = \frac{1}{2} V_m I_m$$

$p=0$ no transformation of energy from electric to non-electric form takes place

$p(t) < 0$, energy extracted from inductor

$p(t) > 0$, energy stored in inductor

(unit: var)

Power for purely capacitive circuit

$$p(t) = -Q \sin 2\omega t \text{ where } \theta_v = \theta_i - 90^\circ$$

$$P = 0$$

$$Q = -\frac{1}{2} V_m I_m$$

$p=0$ average power is zero

$p(t)<0$, energy extracted from capacitor

$p(t)>0$, energy stored in capacitor

(unit: var)

Complex Power

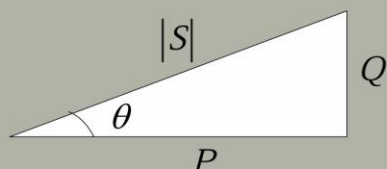
complex power $S = P + jQ$ [VA]

real power P [Watt] $P = R_e \{S\}$

reactive power Q [Var] $Q = I_m \{S\}$ var = volt-ampere reactive

apparent power: $|S| = \sqrt{P^2 + Q^2}$

power factor angle: $\theta = \theta_v - \theta_i$; $\tan \theta = \frac{\sin(\theta_v - \theta_i)}{\cos(\theta_v - \theta_i)} = \frac{Q}{P}$



$Q>0$ inductive lagging current

$Q<0$ capacitive leading current

负载获得最大功率（最大功率传输定理）

第一种情况：负载是纯电阻 (R_L)当负载只能是纯电阻，且电阻值可调时（即负载中没有电感或电容）：最大功率条件：负载电阻等于电源阻抗的模。

$$R_L = |Z_s| = \sqrt{R_s^2 + X_s^2}$$

第二种情况：负载是复阻抗 ($Z_L = R_L + jX_L$)当负载的电阻部分和电抗部分都可以独立调节时：最大功率条件：负载阻抗应与电源阻抗互为共轭复数。

$$Z_L = Z_s^* \Rightarrow R_L = R_s, \quad X_L = -X_s$$

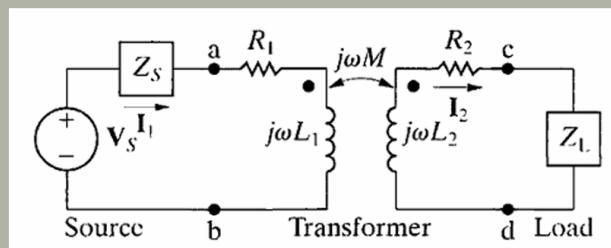
只有当 Z_L 等于 Z_m 的共轭复数时，最大平均功率才能够传输到 Z_L 上，而在有些情况下，这是不可能的。首先， R_L 和 X_L 可能被限制在一定的范围内，这时， R_L 和 X_L 的最优值应是调整 X_L 使其尽可能地接近 $-X_m$ ，同时调整 R_L 使其尽可能地接近 $\sqrt{R_{Th}^2 + (X_L + X_{Th})^2}$ （参见例 10.9）。

另一种情况发生在 Z_L 的幅值可以变化而相位角固定的情况下。此时，只有当 Z_L 的模和 Z_m 的模相等时，最大平均功率才可以传输到负载上，也就是说：

$$|Z_L| = |Z_{Th}| \quad (10.50)$$

Transformer

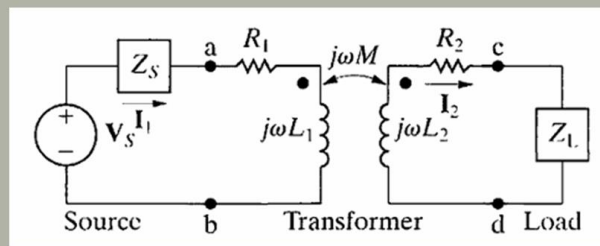
R_1 =resistance of the primary winding
 R_2 =resistance of the secondary winding
 L_1 =self-inductance of the primary winding
 L_2 =self-inductance of the secondary winding
 M =mutual inductance



$V_s = (Z_s + R_1 + j\omega L_1)I_1 - j\omega M I_2$
 $0 = -j\omega M I_1 + (R_2 + j\omega L_2 + Z_L)I_2$
 Let:
 $Z_{11} = Z_s + R_1 + j\omega L_1$
 $Z_{22} = R_2 + j\omega L_2 + Z_L$
 Leads to:

$$I_1 = \frac{Z_{22}}{Z_{11}Z_{22} + \omega^2 M^2} V_s,$$

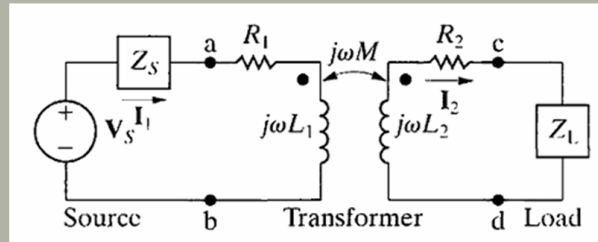
$$I_2 = \frac{j\omega M}{Z_{11}Z_{22} + \omega^2 M^2} V_s = \frac{j\omega M}{Z_{22}} I_1.$$



$$\frac{\mathbf{V}_s}{\mathbf{I}_1} = Z_{\text{int}} = \frac{Z_{11}Z_{22} + \omega^2 M^2}{Z_{22}} = Z_{11} + \frac{\omega^2 M^2}{Z_{22}}.$$

$$Z_{\text{ab}} = Z_{11} + \frac{\omega^2 M^2}{Z_{22}} - Z_s = R_1 + j\omega L_1 + \frac{\omega^2 M^2}{(R_2 + j\omega L_2 + Z_L)}.$$

The last term above called reflected impedance, Z_r .
If the two coils are decoupled, $M=0$ and $Z_{\text{ab}}=R_1+j\omega L_1$



$L_1=L_2$ and coupling coefficient $k=1.0$ $M = k\sqrt{L_1 L_2}$

$$\frac{V_1}{N_1} = \frac{V_2}{N_2}$$

And

$$I_1 N_1 = I_2 N_2$$

But in reality, there is reflected impedance at the transformer load, which makes the equations above having error and hence the turn ratio not being consistent in the lab.

Definition of the Laplace Transform

The Laplace transform of a function is given by the expression:

$$\mathcal{L}\{f(t)\} \equiv F(s) \equiv \int_{0^-}^{\infty} f(t) e^{-st} dt$$

s = Laplace transformation variable, unit of $1/s$

Similar to frequency. Lower limit of the integration is zero.

0^- is to remind to include the origin. $f(t)$ for $t < 0$ is irrelevant.

This defines a “unilateral” or “one-side” Laplace transform.

Cosine: $\mathcal{L}\{\cos(\omega t)\} = \int_{0^-}^{\infty} \cos(\omega t) e^{-st} dt$

$$\begin{aligned} \left[\cos \omega t = \frac{1}{2} (e^{j\omega t} + e^{-j\omega t}) \right] \\ = \int_{0^-}^{\infty} \frac{1}{2} (e^{j\omega t} + e^{-j\omega t}) e^{-st} dt \\ = \int_{0^-}^{\infty} \frac{1}{2} (e^{-(s-j\omega)t} + e^{-(s+j\omega)t}) dt \\ = \frac{1}{2} \left(-\frac{1}{s-j\omega} e^{-(s-j\omega)t} - \frac{1}{s+j\omega} e^{-(s+j\omega)t} \right) \Bigg|_{0^-}^{\infty} \\ = \frac{1}{2} \left(\frac{1}{s-j\omega} + \frac{1}{s+j\omega} \right) = \frac{1}{2} \left(\frac{s+j\omega + s-j\omega}{s^2 + \omega^2} \right) \\ = \frac{s}{s^2 + \omega^2} \end{aligned}$$

sine: check textbook Laplace transform pair

$$\sin \omega t \leftrightarrow \frac{\omega}{s^2 + \omega^2}$$

Addition/subtraction

$$\mathcal{L}\{f_1(t)\} = F_1(s); \mathcal{L}\{f_2(t)\} = F_2(s)$$

$$\mathcal{L}\{f_1(t) \pm f_2(t)\} = \int_{0^-}^{\infty} (f_1(t) \pm f_2(t)) e^{-st} dt$$

$$= \int_{0^-}^{\infty} f_1(t) e^{-st} dt \pm \int_{0^-}^{\infty} f_2(t) e^{-st} dt = F_1(s) \pm F_2(s)$$

Laplace Transform /Operational Transform

Integration:

Using integral by parts $\int u dv = uv - \int v du$

$$\mathcal{L}\left\{\int_{0^-}^t f(x) dx\right\} = \int_{0^-}^{\infty} \left[\int_{0^-}^t f(x) dx\right] e^{-st} dt$$

We can do it by integration by parts or using Laplace

Transform Derivative results.

$$u = \int_{0^-}^t f(x) dx; dv = e^{-st} dt; du = f(t) dt; v = -\frac{e^{-st}}{s} \Bigg|_{0^-}^{\infty}$$

$$\text{w.r. to time, } t \leftarrow \left[-\frac{e^{-st}}{s} \int_{0^-}^t f(x) dx \right] + \int_{0^-}^{\infty} \frac{e^{-st}}{s} f(t) dt = 0 + \frac{F(s)}{s} = \frac{F(s)}{s}$$

$$\text{Alternate proof: } \mathcal{L}\left\{\int_{0^-}^t f(x) dx\right\} = \mathcal{L}\{g(t)\}; \frac{dg}{dt} = f(t); g(0) = 0$$

$$\mathcal{L}\left\{\frac{dg}{dt} f(t)\right\} = sF(s) - f(0^-) \quad \mathcal{L}\left\{\frac{dg}{dt}\right\} = \mathcal{L}\{f(t)\}; sG(s) = F(s); G(s) = \frac{F(s)}{s}$$

$$G(s) = \mathcal{L}\left\{\int_{0^-}^t f(x) dx\right\} = \frac{F(s)}{s}$$

$$I_c = C \frac{dV_c}{dt}$$

$$I_c = CS V_c(S)$$

$$\frac{V_c(s)}{I_c(s)} = \frac{1}{CS}$$

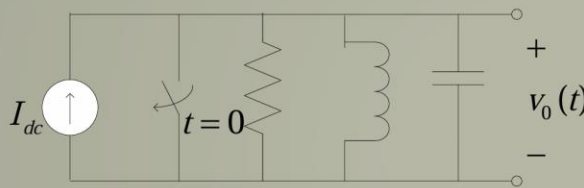
$$V_L = L \frac{dI_L}{dt}$$

$$V_L = LS I_L(s)$$

$$\frac{V_L}{I_L} = LS$$

Applying the Laplace Transform

Use this circuit as an example: The differential equations of current and voltage



$$L \left\{ \frac{d}{dt} f(t) \right\} = sF(s) - f(0^-)$$

$$G(s) = L \left\{ \int_{0^-}^t f(x) dx \right\} = \frac{F(s)}{s}$$

$$v(t) = i_R(t) R = L \frac{di_L(t)}{dt} = \int C \frac{dv_c}{dt} dt$$

$$i(t) = \frac{v_o(t)}{R} + \frac{1}{L} \int_0^t v(x) dx + C \frac{dv_o(t)}{dt} = I_{dc} u(t)$$

This yields:

$$\frac{I_{dc}}{s} = \frac{V(s)}{R} + \frac{1}{L} \frac{V(s)}{s} + CsV(s)$$

Inductor => Ls

$$\frac{I_{dc}}{s} = V(s) \left[\frac{1}{R} + \frac{1}{Ls} + Cs \right]$$

$$\Rightarrow V(s) = \frac{I_{dc} \frac{1}{C}}{s \left(\frac{1}{R} + \frac{1}{Ls} + Cs \right) \frac{1}{C}} = \frac{I_{dc} / C}{s^2 + \frac{1}{RC} s + \frac{1}{LC}}$$

Laplace transform converts ordinary differential equations into functions of frequency-like (s) algebraic equations which are much simpler to solve.

12.7.1 部分分式展开法：有理真分式

对每个 $D(s)$ 的根，有理真分式可以展开为部分分式之和的形式。在进行部分分式展开之前，应对 $D(s)$ 进行因式分解，对每个 $D(s)$ 的不等根，部分分式之和中都会出现一项，对 $D(s)$ 的 r 阶重根，展开式中就会含有 r 项。例如，在下面的真分式中

$$\frac{s+6}{s(s+3)(s+1)^2}$$

分母有 4 个根，两个独立的根，即 $s=0$, $s=-3$ ，一个二阶重根出现在 $s=-1$ 处，这样，其展开式的形式为

$$\frac{s+6}{s(s+3)(s+1)^2} \equiv \frac{K_1}{s} + \frac{K_2}{s+3} + \frac{K_3}{(s+1)^2} + \frac{K_4}{s+1} \quad (12.43)$$

部分分式法的关键就是要找到展开式中每一项所对应的时域函数 $f(t)$ ，根据表 12.1 可以得到

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{s+6}{s(s+3)(s+1)^2} \right\} \\ = (K_1 + K_2 e^{-3t} + K_3 t e^{-t} + K_4 e^{-t}) u(t) \end{aligned} \quad (12.44)$$

剩下的问题就是确定部分分式展开式中的系数 ($K_1, K_2, K_3, K_4, \dots$)。 $D(s)$ 的根会有 4 种可能的情况：(1)实数的单根；(2)复数的单根；(3)实数的重根；(4)复数的重根。在顺序讨论每种情况之前，先说明以下几点。

$$F(s) = \frac{100 \times (s + 3)}{(s + 6)(s^2 + 6s + 25)} \quad (12.51)$$

注意到 $F(s)$ 是有理真分式，下面求二次项 $s^2 + 6s + 25$ 的根

$$s^2 + 6s + 25 = (s + 3 - j4)(s + 3 + j4) \quad (12.52)$$

对分母因式分解后

$$\begin{aligned} \frac{100 \times (s + 3)}{(s + 6)(s^2 + 6s + 25)} &\equiv \\ \frac{K_1}{s + 6} + \frac{K_2}{s + 3 - j4} + \frac{K_3}{s + 3 + j4} \end{aligned} \quad (12.53)$$

用同样的方法求 K_1, K_2, K_3 :

$$K_1 = \left. \frac{100 \times (s + 3)}{s^2 + 6s + 25} \right|_{s=-6} = \frac{100 \times (-3)}{25} = -12 \quad (12.54)$$

$$\begin{aligned} K_2 &= \left. \frac{100 \times (s + 3)}{(s + 6)(s + 3 + j4)} \right|_{s=-3-j4} = \frac{100 \times (j4)}{(3 + j4)(j8)} \\ &= 6 - j8 = 10e^{-j53.13^\circ} \end{aligned} \quad (12.55)$$

$$\begin{aligned} K_3 &= \left. \frac{100 \times (s + 3)}{(s + 6)(s + 3 - j4)} \right|_{s=-3-j4} = \frac{100 \times (-j4)}{(3 - j4)(-j8)} \\ &= 6 + j8 = 10e^{j53.13^\circ} \end{aligned} \quad (12.56)$$

得:

$$\begin{aligned} \frac{100 \times (s + 3)}{(s + 6)(s^2 + 6s + 25)} &= \frac{-12}{s + 6} + \frac{10 \angle -53.13^\circ}{s + 3 - j4} \\ &\quad + \frac{10 \angle 53.13^\circ}{s + 3 + j4} \end{aligned} \quad (12.57)$$

需要说明的是，第一，在物理可实现的电路中，复根总是以共轭复数对出现的；第二，与共轭复数对相对应的系数也是共轭复数，例如， K_3 [参见式 (12.56)] 就是 K_2 [参见式 (12.55)] 的共轭复数，因此，对于共轭复根，实际上只需要计算一半参数。

在对方程 (12.57) 进行反变换之前，要验算部分分式展开式，选 -3 时验算比较方便，因为此时左边为零

$$\begin{aligned} F(s) &= \frac{-12}{3} + \frac{10 \angle -53.13^\circ}{-j4} + \frac{10 \angle 53.13^\circ}{j4} \\ &= -4 + 2.5 \angle 36.87^\circ + 2.5 \angle -36.87^\circ \\ &= -4 + 2.0 + j1.5 + 2.0 - j1.5 = 0 \end{aligned}$$

现在进行反变换:

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{100 \times (s + 3)}{(s + 6)(s^2 + 6s + 25)} \right\} &= (-12e^{-6t} + 10e^{-j53.13^\circ} e^{-(3-j4)t} \\ &\quad + 10e^{j53.13^\circ} e^{-(3+j4)t})u(t) \end{aligned} \quad (12.58)$$

一般来说，使函数在时域中包含虚部是不现实的，而恰好含有虚部的项总是成对出现，因此可以将出现的虚数相加并消掉

Inverse Laplace Transform

CASE 2: DISTINCT COMPLEX ROOTS

Use the pair:

$$e^{-at}u(t) \leftrightarrow \frac{1}{s+a}$$

$$\frac{K}{s+\alpha-j\beta} + \frac{K^*}{s+\alpha+j\beta} \leftrightarrow \left[Ke^{-(\alpha-j\beta)t} + K^* e^{-(\alpha+j\beta)t} \right] u(t)$$

Let $K = |K| \angle \theta; K^* = |K| \angle -\theta$

The time function becomes:

$$|K| \left[e^{-\alpha t} e^{j\beta t} e^{j\theta} + e^{-\alpha t} e^{-j\beta t} e^{-j\theta} \right] u(t)$$

Use: $\cos x = \frac{e^{jx} + e^{-jx}}{2}$

$$\Rightarrow \left[2|K| e^{-\alpha t} \cos(\beta t + \theta) \right] u(t) \text{ which is useful as a general transform pair}$$

CASE 3: REPEATED REAL ROOTS

Thus, we obtain:

$$F(s) = \frac{1}{s} + \frac{3}{s+1} + \frac{2}{(s+1)^2}$$

Transform pair:

$$\mathcal{L}\{te^{-at}\} \leftrightarrow \frac{1}{(s+a)^2}$$

$$\mathcal{L}\{e^{-at}\} \leftrightarrow \frac{1}{s+a}$$

The inverse Laplace transform is then?

$$f(t) = \left[1 + 3e^{-t} + 2te^{-t} \right] u(t)$$

The transfer function of a circuit will have the form

$$F(s) = \frac{K(s+Z_1)(s+Z_2) \times \dots \times (s+Z_n)}{(s+P_1)(s+P_2) \times \dots \times (s+P_n)}$$

The poles are the roots of the denominator polynomial

$$-P_1, -P_2, -P_3, \dots, -P_n$$

The zeros are the roots of the numerator polynomial

$$-Z_1, -Z_2, -Z_3, \dots, -Z_n$$

零点就是分子联立方程后的根

极点就是分母联立方程后的根

Initial and Final Value Theorems

The initial- and final-value theorems are useful because they enable us to determine from $F(s)$ the behavior of $f(t)$ at 0 and ∞

$$\text{Initial value: } \lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} sF(s)$$

$$\text{Final value: } \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

我们要证明: $\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} sF(s)$ 。

$$\int_{0^-}^{\infty} f'(t)e^{-st} dt = sF(s) - f(0^-)$$

第二步: 取极限 $s \rightarrow \infty$

当 $s \rightarrow \infty$ 时, 看左边的积分项。整个指数项 e^{-st} 就会变成 0。

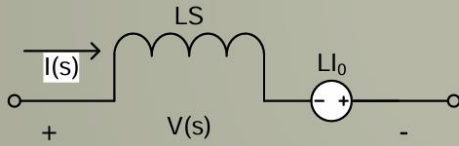
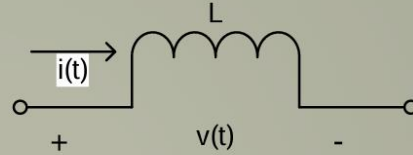
$$\lim_{s \rightarrow \infty} \int_{0^-}^{\infty} f'(t)e^{-st} dt = 0$$

$$0 = \lim_{s \rightarrow \infty} [sF(s) - f(0^-)]$$

$$\lim_{s \rightarrow \infty} sF(s) = f(0^-)$$

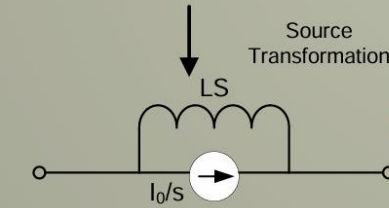
Inductor

$$v(t) = L \frac{di(t)}{dt}$$



Take Laplace transform

$$\begin{aligned} V(s) &= \mathcal{L}\left\{L \frac{di(t)}{dt}\right\} = L(sI(s) - i(0^+)) \\ &= LsI(s) - LI_0 \text{ (voltage)} \end{aligned}$$



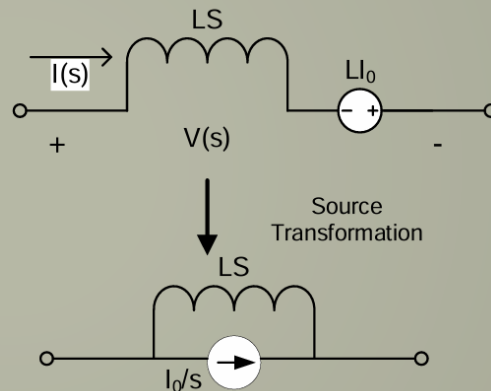
LI_0 = the initial condition creates the equivalent of an additional source

$$I = V/R \Rightarrow I = LI_0/Ls = I_0/s$$

S-domain

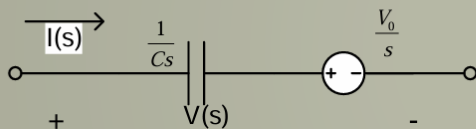
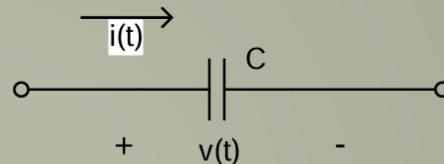
$$V(s) = sLI(s) - LI_0$$

$$I(s) = \frac{V(s)}{Ls} + \frac{I_0}{s}$$



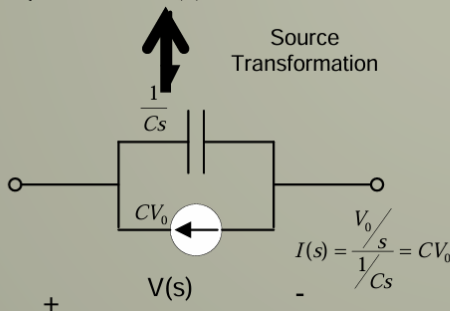
Capacitor:

$$i(t) = C \frac{dv(t)}{dt}$$



Take Laplace transform

$$\begin{aligned} I(s) &= \mathcal{L}\left\{C \frac{dv(t)}{dt}\right\} = C(sV(s) - v(0^+)) \\ &= CsV(s) - CV_0 \text{ (voltage)} \end{aligned}$$



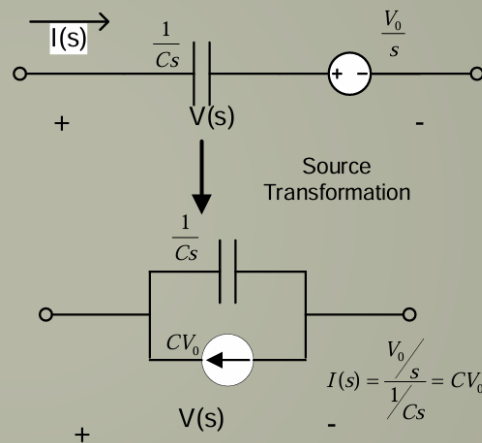
CV_0 = the initial condition creates the equivalent of an additional source

Capacitor

S-domain

$$I(s) = CsV(s) - CV_0$$

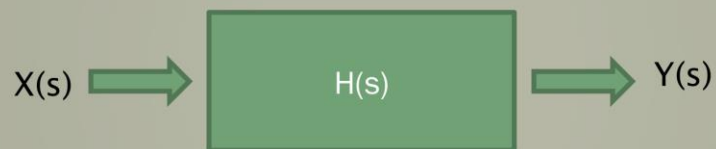
$$V(s) = \frac{I(s)}{Cs} + \frac{V_0}{s}$$



Transfer Function

$$H(s) = \frac{Y(s)}{X(s)}$$

$Y(s)$: output signal $\rightarrow$ response;
 $X(s)$ input signal $\rightarrow$ source



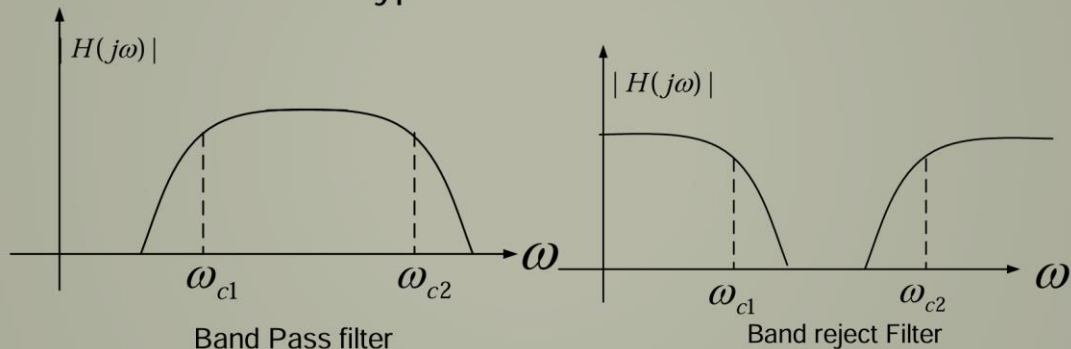
$$Y(s) = H(s) X(s)$$

Poles of $H(s)$ give transient response
 Poles of $X(s)$ give ss response

Some preliminaries

Filters are frequency selective circuits

Several common types



Cut-off frequency ω_c is usually defined at half-power point,
 i.e. where the response has fallen by a factor of $\sqrt{2}$

- The frequency of the Bode plots are plotted against a logarithmic frequency axis. The Bode magnitude plot measures the system Output/input ratio in special unit called decibel (dB).

$$dB = 20 \log_{10} \left| \frac{V_{out}}{V_{in}} \right|$$

Bode Plot (Bode Magnitude Plots)

- Step 1: Solve zeros and poles of the transfer function.
- Step 2: Use frequency=0 as the starting point, and convert the magnitude into decibel.
- The locations of every magnitude pole and Zeros are break frequencies. At Zero break frequency, the slope of the line increases by 20dB/decade per zero. At a pole, the slope of the line decreases by 20dB/decade per pole.
- At a zero break frequency, the actual value of the pole increases from the straight line plot by 3 dB, and pole is 3 dB decreases.

Bode Plot (Bode Phase Plots)

- For every zero, slope the line up $45^\circ/\text{decade}$, while ω is one decade before the break frequency. Flatten the slope when the phase change by 90° .
- For every pole, slope the line down $45^\circ/\text{decade}$, while ω is one decade before the break frequency. Flatten the slope when the phase change by -90° .

感抗 (Inductive Reactance) $X_L = \omega L = 2\pi fL$

容抗 (Capacitive Reactance) $X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$

电抗 (Reactance) $X = X_L - X_C = \omega L - \frac{1}{\omega C}$

阻抗 (Impedance) $Z = R + jX$

品质因数 (5个特征参数的最后一个) 定义为中心频率与带宽之比, 利用式 (14.25) 和式 (14.32) 得

$$\begin{aligned} Q &= \omega_0 / \beta \\ &= \frac{(1/LC)}{(R/L)} \\ &= \sqrt{\frac{L}{CR^2}} \end{aligned} \quad (14.33)$$

现在, 已得到了串联RLC电路谐振的5个特征参数: (1) 两个截止频率 ω_{c1} 和 ω_{c2} , 确定了通